

PRATEEK SAXENA

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SUMMARY

AI Engineer building LLM-powered applications and AI automation tools, backed by 7+ years of experience across marketing analytics, analytics engineering, and business data systems. Experienced in developing FastAPI, React, OpenAI API, LangChain, SQL, and Python-based solutions that turn business workflows, customer data, and internal knowledge into usable AI products.

SKILLS

Languages	Python, SQL, JavaScript, TypeScript
Frameworks	FastAPI, React, Next.js, Node.js, REST APIs
GenAI	LangChain, LangGraph, LLM Applications, Multi-Agent Systems, Tool Calling, Structured Outputs, Prompt Engineering, Long-Term Memory
RAG	Advanced RAG, Corrective RAG, Self-RAG, Embeddings, Vector Databases, pgvector, Semantic Search, Metadata Filtering, Retrieval Evaluation
Machine Learning	Scikit-learn, Pandas, NumPy, Regression, Classification, Clustering, Churn Prediction, Customer Segmentation, Time Series Forecasting, PyTorch Basics
Databases	PostgreSQL, SQL Server, Advanced SQL, ETL Pipelines, Data Validation, Star Schema
Analytics	Power BI, DAX, Excel, KPI Design, Cohort Analysis, A/B Testing, CAC, ROAS, LTV, CLV, RFM Modeling
Cloud / LLMOps	LangSmith, AWS S3, AWS EC2, Docker, Git, CI/CD, Logging, Monitoring

EXPERIENCE

AI Engineer Growith	May 2023 – Present Dallas, TX
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Corrective RAG Knowledge Support Agent

- Built a Corrective RAG knowledge support agent for Growith’s 10+ brand ecosystem, ingesting product docs, pricing rules, FAQs, feature workflows, support playbooks, objection-handling content, and vertical-specific use cases from AWS S3 and internal repositories into a searchable AI knowledge layer for sales and customer support teams.
- Designed a LangGraph-based retrieval pipeline using PostgreSQL + pgvector, semantic search, metadata filtering, query rewriting, document relevance grading, fallback retrieval, and grounded answer generation to correct weak retrieval before answering product, pricing, troubleshooting, and implementation questions.
- Deployed the FastAPI-based service with Docker on AWS, reducing sales-to-technical clarification tickets by **42%** and customer-support escalations for product workflow and “not working as expected” queries by **56%** over 3 months through consistent, source-backed answers.

Multi-Agent AI Business Analyst Copilot

- Built a multi-agent AI business analyst copilot for franchise operators, including Chicken Express, Teriyaki Madness, and Donatos Pizza locations, translating natural-language questions into validated SQL queries, Python-based calculations, root-cause analysis, and executive-ready insights across sales, customer, campaign, location, and product performance data.
- Orchestrated a LangGraph-based agent workflow with intent classification, schema selection, SQL generation, data quality validation, metric computation, insight synthesis, response verification, and LangSmith observability to trace agent behavior, debug failed runs, and prevent hallucinated business conclusions.

- Implemented long-term memory with PostgreSQL-backed conversation history and vector-based semantic retrieval to preserve franchise-specific context, KPI definitions, prior analysis patterns, and stakeholder preferences, cutting insight turnaround from hours to minutes while reducing manual reporting workload by **64%** and ad-hoc analyst requests by **47%**.

Tool-Calling AI Voice Agent Platform

- Built a tool-calling AI voice agent platform for restaurant, retail, and service-business clients, enabling inbound callers to check business hours, ask service/product questions, request bookings, place order inquiries, qualify leads, and receive SMS follow-ups through real-time conversational workflows.
- Designed the agent architecture with speech-to-text, LLM reasoning, function calling, structured outputs, CRM/calendar/order-management API integrations, human escalation logic, call transcript storage, and post-call summaries to convert unstructured phone conversations into trackable business actions.
- Reduced missed-call leakage by **67%** and manual front-desk follow-up by **39%** by allowing the AI receptionist to answer, qualify, route, summarize, and trigger next-step workflows without requiring staff involvement on every inbound call.

ML-Powered Customer Segmentation & Churn Intelligence Platform

- Built an ML-powered customer segmentation and churn intelligence platform for restaurant and retail clients, classifying customers into high-value, at-risk, low-value, lapsed, repeat, and win-back segments to support retention, campaign targeting, and revenue optimization.
- Developed end-to-end model pipelines with SQL-based feature engineering, RFM analysis, cohort behavior, purchase frequency, recency, average order value, logistic regression/tree-based churn scoring, and clustering models, with FastAPI endpoints serving predictions to a React-based dashboard.
- Improved at-risk customer identification by **37%** and reduced manual segmentation work by **51%**, enabling teams to trigger targeted win-back campaigns, send personalized discount offers to churn-prone customers, and improve retention by prioritizing outreach before customers became inactive.

Analytics Engineer

Jan 2020 – Jul 2022

Rapid Punching

India

- Rebuilt revenue reporting models in SQL and Power BI to separate gross sales, discounts, refunds, shipping, taxes, and net revenue, giving leadership a more accurate view of weekly performance and helping identify margin leakage from high-discount products and refund-heavy SKUs.
- Designed modular SQL transformation workflows across raw order, customer, product, refund, and campaign data, creating cleaner reporting tables for sales, retention, marketing, and product analysis while reducing repeated manual data prep in Excel by nearly **30%**.
- Developed Python and pandas-based data quality checks to flag duplicate orders, missing customer IDs, SKU mismatches, refund inconsistencies, and broken UTM parameters before reports were refreshed, improving trust in Power BI dashboards used for revenue, ROAS, and product performance reviews.
- Analyzed first-to-second purchase drop-off, recency, frequency, AOV, and repeat purchase behavior using SQL, Power BI, and Excel, helping marketing teams build more practical segments for lapsed buyers, high-value customers, and discount-sensitive shoppers.
- Created product-level sales forecasting models in Python using historical order data, seasonality, recent sales velocity, and promotional periods, improving demand visibility for fast-moving products and helping merchandising teams reduce overstock and stockout blind spots during campaign periods.

Data Analyst

Jan 2018 – Nov 2019

Web Ignito

India

- Analyzed marketing, sales, and customer behavior data for agency clients across eCommerce, SaaS, and local service industries, using SQL, Excel, and Power BI to track CAC, ROAS, conversion rates, lead quality, funnel drop-offs, and customer acquisition trends.

- Built recurring Power BI dashboards and cleaned multi-source datasets from Google Analytics, Meta Ads, Google Ads, CRM exports, and website forms, reducing manual reporting effort by **38%** and improving visibility into campaign performance, channel efficiency, and customer journey bottlenecks.
- Delivered weekly data-backed recommendations on budget allocation, landing page performance, audience targeting, and offer strategy, helping clients identify underperforming campaigns, improve lead quality, and make faster marketing decisions from performance insights.

EDUCATION

The University of Texas at Dallas	May 2024
Master of Science, Marketing	GPA: 3.97
<i>Focus: Customer Analytics, Predictive Modeling, Advanced Web Analytics</i>	
Vishwakarma Institute of Technology (VIT)	Pune, India
Bachelor of Technology, Chemical Engineering	